

A Prime–Resolvent Research Programme for Spectral Approximations to the Riemann Ξ Function

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Status and scope. This document does not claim a proof of the Riemann hypothesis. It isolates an abstract Stieltjes/slit-plane criterion, an explicit prime-tail target, and a three-part error budget for spectral approximants. The concrete spectral alignment and full convergence remain open. Some rate calculations are labelled candidate results pending a line-by-line source and convention audit.

Abstract

Recent work constructs self-adjoint spectral approximants from Euler products over primes up to λ^2 and reports striking numerical agreement with low non-trivial zeros of the Riemann zeta function. A rigorous convergence theorem would imply the Riemann hypothesis. We formulate a research programme based on the positive observable

$$\frac{1}{2} \operatorname{Tr}(D^2 + xI)^{-1},$$

rather than direct convergence of regularized determinants. The central abstract observation is that a holomorphic extension to the slit plane of the critical-line logarithmic-derivative target excludes non-real zeros of Ξ . Positive Stieltjes approximants with one-point control provide such extensions through normal-family compactness. On the arithmetic side, the target admits an absolutely convergent prime expansion for $x > 1/4$, with an elementary explicit tail majorant. We separate the desired comparison into spectral, model, and prime errors, describe a certificate-driven finite-dimensional route, and give a Lean 4 formalization plan. The accompanying repository contains an axiom-free elementary Lean seed and a strict publication gate.

1 Introduction

Let

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad \Xi(z) = \xi\left(\frac{1}{2} + iz\right).$$

The Riemann hypothesis is equivalent to all zeros of the entire even function Ξ being real. The Hilbert–Pólya philosophy seeks a self-adjoint operator whose spectrum encodes those zeros. Connes, Consani and Moscovici recently proposed self-adjoint operators obtained from rank-one perturbations of a scaling spectral triple on $[\lambda^{-1}, \lambda]$; their construction uses Euler products over primes $p \leq \lambda^2$, and numerical experiments show high-precision agreement with low Riemann zeros. They state explicitly that a rigorous proof of convergence would establish RH and discuss regularized determinants as approximants to Ξ [1].

A companion theorem of Connes and van Suijlekom shows, under lower-bounded self-adjointness and a simple isolated lowest eigenvalue with an even eigenfunction, that the Fourier transform of

that eigenfunction has only real zeros [2]. This supplies a real-zero mechanism for each finite or semilocal model but does not prove convergence to Ξ .

The present programme proposes to compare squared resolvents instead of entire determinants. Resolvents are positive, suppress high spectral values, and are stable under perturbation. More importantly, the corresponding scalar functions lie in the Stieltjes–Nevanlinna class, where local boundedness and normal-family arguments are natural.

The contribution of this draft is organizational and reductive. It gives a complete proof of an abstract slit-plane extension criterion, isolates an explicit arithmetic target and prime tail, and states precisely what remains to be shown for the concrete operators. It should be read as a blueprint for formalization and further analysis, not as a completed RH argument.

2 The critical-line resolvent target

For $x > 1/4$, define

$$\mathcal{S}_\Xi(x) = \frac{1}{2\sqrt{x}} \frac{\xi'}{\xi} \left(\frac{1}{2} + \sqrt{x} \right). \quad (1)$$

The restriction $x > 1/4$ ensures that $\sigma = 1/2 + \sqrt{x} > 1$, so the Euler product is absolutely convergent and $\zeta(\sigma) \neq 0$.

If RH holds and the positive zeros of Ξ are γ_k , counted with multiplicity, the symmetric canonical product formally gives

$$\mathcal{S}_\Xi(x) = \sum_{k \geq 1} \frac{1}{\gamma_k^2 + x}. \quad (2)$$

Thus $\mathcal{S}_\Xi$ is a Stieltjes transform of a positive discrete measure. A final paper must verify the canonical-product normalization carefully. The one-way criterion below does not assume (2).

3 A slit-plane extension criterion

Let

$$\Omega = \mathbb{C} \setminus (-\infty, 0].$$

Theorem 3.1 (Slit-plane extension criterion). *Suppose there is a holomorphic function $S : \Omega \rightarrow \mathbb{C}$ and a nonempty open interval $I \subset (1/4, \infty)$ such that*

$$S(x) = \mathcal{S}_\Xi(x) \quad (x \in I).$$

Then all zeros of Ξ are real.

Proof. For z in the open upper half-plane define

$$M(z) = 2zS(-z^2).$$

The map $z \mapsto -z^2$ takes the upper half-plane into Ω . Indeed, if $\operatorname{Re} z = 0$, then $-z^2 > 0$; otherwise $\operatorname{Im}(-z^2) = -2\operatorname{Re} z \operatorname{Im} z \neq 0$. Hence M is holomorphic.

Take $z = iy$ with $y > 0$ and $y^2 \in I$. The functional equation $\xi(s) = \xi(1-s)$ gives $\xi'(s) = -\xi'(1-s)$, and therefore

$$-\frac{\Xi'(iy)}{\Xi(iy)} = i \frac{\xi'}{\xi} \left(\frac{1}{2} + y \right) = 2iyS(y^2) = M(iy).$$

The function $-\Xi'/\Xi$ is holomorphic away from the discrete zeros of Ξ . The upper half-plane minus a discrete subset is connected, so the identity theorem yields

$$M(z) = -\frac{\Xi'(z)}{\Xi(z)}$$

away from the zeros.

If z_0 were an upper-half-plane zero of multiplicity $m > 0$, local factorization would give $\Xi(z) = (z - z_0)^m g(z)$ with $g(z_0) \neq 0$. Hence

$$-\frac{\Xi'(z)}{\Xi(z)} = -\frac{m}{z - z_0} - \frac{g'(z)}{g(z)},$$

which has a nonremovable pole at z_0 , contradicting holomorphy of M . Conjugation symmetry excludes zeros in the lower half-plane. Thus every zero is real. $\square$

Remark 3.2. The theorem only requires an extension on the slit plane agreeing with the target on one interval. It does not require prior identification of every zero, global determinant convergence, or an explicit inverse spectral theorem.

4 Stieltjes approximants and compactness

Let ν_j be positive measures on $[0, \infty)$ and define

$$S_j(w) = \int_0^\infty \frac{d\nu_j(t)}{t + w}, \quad w \in \Omega.$$

For a discrete paired spectrum $\{\pm\gamma_{j,k}\}$ one expects

$$S_j(x) = \sum_k \frac{1}{\gamma_{j,k}^2 + x} = \frac{1}{2} \text{Tr}(D_j^2 + xI)^{-1}.$$

Proposition 4.1 (One-point control). *Fix $x_0 > 0$. If $\sup_j S_j(x_0) < \infty$, then the family $\{S_j\}$ is locally bounded on Ω .*

Proof. For each compact $K \subset \Omega$, continuity and avoidance of the negative axis imply the existence of C_K with

$$\frac{1}{|t + w|} \leq \frac{C_K}{t + x_0}$$

for all $t \geq 0$ and $w \in K$. Integrating gives $|S_j(w)| \leq C_K S_j(x_0)$. $\square$

Montel's theorem then gives compact-open convergent subsequences. If $S_j(x)$ converges to $\mathcal{S}_\Xi(x)$ on an open interval, any subsequential limit supplies the extension required by Theorem 3.1.

Theorem 4.2 (Abstract prime-resolvent criterion). *Assume S_j are Stieltjes transforms of positive measures, are uniformly bounded at one positive point, and converge pointwise to $\mathcal{S}_\Xi$ on a nonempty open interval contained in $(1/4, \infty)$. Then RH holds.*

Proof. Combine Proposition 4.1, Montel compactness, the identity theorem, and Theorem 3.1. $\square$

5 The arithmetic side

For real $\sigma > 1$,

$$\frac{\xi'}{\xi}(\sigma) = \frac{1}{\sigma} + \frac{1}{\sigma-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(\sigma/2) - \sum_{n \geq 2} \frac{\Lambda(n)}{n^\sigma}, \quad (3)$$

where $\psi = \Gamma'/\Gamma$ and Λ is the von Mangoldt function. Putting $y = \sqrt{x}$ and $\sigma = 1/2 + y$, define the truncation

$$P_X(x) = \frac{1}{2y} \left[\frac{1}{\sigma} + \frac{1}{\sigma-1} - \frac{1}{2} \log \pi + \frac{1}{2} \psi(\sigma/2) - \sum_{2 \leq n \leq X} \frac{\Lambda(n)}{n^\sigma} \right]. \quad (4)$$

Fix $\delta > 0$ and suppose $\sigma \geq 1 + \delta$. Since $0 \leq \Lambda(n) \leq \log n$, a decreasing-integral comparison for $X \geq 3$ gives

$$\sum_{n > X} \frac{\Lambda(n)}{n^\sigma} \leq X^{-\delta} \left(\frac{\log X}{\delta} + \frac{1}{\delta^2} \right). \quad (5)$$

This produces a uniform explicit bound for $|\mathcal{S}_\Xi(x) - P_X(x)|$ on any compact interval corresponding to $y \geq 1/2 + \delta$.

The Lean seed formalizes the closed-form majorant and its sign. The full sum–integral comparison and log-derivative formula are formalization targets.

6 The three-part error budget

Let $S_{\lambda,N}$ be a concrete spectral observable and $\tilde{S}_\lambda$ a source-audited model observable. Then

$$|S_{\lambda,N} - P_{\lambda^2}| \leq |S_{\lambda,N} - \tilde{S}_\lambda| + |\tilde{S}_\lambda - \mathcal{S}_\Xi| + |\mathcal{S}_\Xi - P_{\lambda^2}| \quad (6)$$

$$=: E_{\text{spectral}} + E_{\text{model}} + E_{\text{prime}}. \quad (7)$$

This elementary inequality and addition of component limits are already encoded in Lean.

The arithmetic term is controlled by (5). The model term depends on exact Fourier/Mellin conventions and is presently under source audit. The spectral term is the decisive open contribution.

7 Spectral alignment and finite certificates

Let A be a finite Hermitian matrix with lowest eigenvalues $\epsilon_0 < \epsilon_1$ and gap $g = \epsilon_1 - \epsilon_0$. For a normalized trial vector v define

$$r = \langle Av, v \rangle - \epsilon_0.$$

Expansion in an eigenbasis yields

$$\inf_{\theta} \|v - e^{i\theta} \xi_0\| \leq \sqrt{2r/g}.$$

A Galerkin projection tail τ then motivates the scalar defect

$$\eta_{\lambda,N} = \|k_\lambda\| \sqrt{2r_{\lambda,N}/g_{\lambda,N} + 2\tau_{\lambda,N}}.$$

For computation, exact knowledge of ϵ_0 should be replaced by a residual and a certified separation bound. The external program exports rational or interval data; Lean checks Hermitian symmetry, residual inequalities, gap/separation conditions, and the resulting scalar bound.

No numerical result is accepted as proof without a certificate and a theorem connecting the certificate to the infinite model.

8 Candidate rate and its status

An exploratory calculation suggested that a bound

$$\eta_{\lambda, N(\lambda)} = O(\lambda^{-q}), \quad q > 1/2,$$

may be enough to force total convergence on a suitable interval, with optimized exponent

$$\rho(q) = \min \left\{ \frac{2q-1}{3}, \frac{2}{3} \right\}.$$

The scalar positivity and saturation properties of this expression are included in Lean. However, the analytic derivation of the 2/3 ceiling depends on a candidate enlargement of a prolate/model estimate. Before this can be stated as a theorem, one must verify the precise transform normalization, support, constants, and source hypotheses in [1]. Accordingly this section is a research target, not a claimed result.

9 Lean formalization architecture

Mathlib supplies definitions of the Riemann zeta function, a pole-removed completed zeta, a functional equation, and a proposition named ‘RiemannHypothesis’. The formalization is divided into:

1. the exact ξ/Ξ bridge;
2. geometry of the slit plane and the map $z \mapsto -z^2$;
3. local analytic factorization and logarithmic-derivative poles;
4. the extension criterion;
5. the explicit prime tail;
6. finite Stieltjes and spectral certificates;
7. normal-family compactness;
8. the concrete unbounded operator.

The seed repository contains no project axioms and represents open results as propositions or explicit fields of a future package. The operator layer will not be introduced until the abstract analytic chain is complete.

10 Publication criteria

A first formalization paper becomes defensible after the abstract slit-plane criterion, the Mathlib RH bridge, the prime-tail theorem, and the finite certificate layer are all complete and reproducible. A substantive analytic paper additionally needs a new concrete estimate for the Galerkin, gap, or state-alignment error. A claim of RH requires full convergence, absence of circular use of Weil positivity, and independent specialist review.

11 Conclusion

The squared-resolvent/Stieltjes viewpoint converts an imprecise aspiration (“the spectral determinant converges to Ξ ”) into several separable and falsifiable tasks. The extension criterion shows that convergence on one real interval can be enough, while positivity provides compactness and the Euler product provides an explicit arithmetic target. The remaining hard step is not hidden: it is a quantitative, non-circular comparison between the concrete lowest spectral state and the target model, together with all trace and domain properties needed to pass to the limit.

References

- [1] A. Connes, C. Consani, and H. Moscovici, *Zeta Spectral Triples*, arXiv:2511.22755, 2025.
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- [4] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, 1859.